CORT-Blerd

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Data Set Up

```r
# 1.
library(tidyverse)
```

```
Warning: package 'ggplot2' was built under R version 4.4.3
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✔ dplyr     1.1.4     ✔ readr     2.1.5
✔ forcats   1.0.0     ✔ stringr   1.5.1
✔ ggplot2   4.0.1     ✔ tibble    3.2.1
✔ lubridate 1.9.4     ✔ tidyr     1.3.1
✔ purrr     1.0.4
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to b
```

```r
library(car)
```

```
Loading required package: carData

Attaching package: 'car'

The following object is masked from 'package:dplyr':

    recode

The following object is masked from 'package:purrr':

    some
```

```r
library(gt)
library(modelsummary)
library(emmeans)
```

```
Welcome to emmeans.
Caution: You lose important information if you filter this package's results.
See '? untidy'
```

```r
library(modelbased)
```

```
Warning: package 'modelbased' was built under R version 4.4.3
```

```r
library(multcomp)
```

```
Loading required package: mvtnorm

Attaching package: 'mvtnorm'

The following object is masked from 'package:modelbased':

    standardize

Loading required package: survival
Loading required package: TH.data
Loading required package: MASS

Attaching package: 'MASS'

The following object is masked from 'package:dplyr':

    select


Attaching package: 'TH.data'

The following object is masked from 'package:MASS':

    geyser
```

```r
library(ggpubr)
library(cowplot)
```

```

Attaching package: 'cowplot'

The following object is masked from 'package:ggpubr':

    get_legend

The following object is masked from 'package:gt':

    as_gtable

The following object is masked from 'package:lubridate':

    stamp
```

```r
library(lme4)
```

```
Warning: package 'lme4' was built under R version 4.4.3

Loading required package: Matrix

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

    expand, pack, unpack
```

```r
library(lmerTest)
```

```

Attaching package: 'lmerTest'

The following object is masked from 'package:lme4':

    lmer

The following object is masked from 'package:stats':

    step
```

```r
library(outliers)
library(DHARMa)
```

```
This is DHARMa 0.4.7. For overview type '?DHARMa'. For recent changes, type news(package
```

```r
library(statmod)
library(readr)
library(dplyr)
library(EnvStats)
```

```

Attaching package: 'EnvStats'

The following object is masked from 'package:Matrix':

    print

The following object is masked from 'package:MASS':

    boxcox

The following object is masked from 'package:car':

    qqPlot

The following objects are masked from 'package:stats':

    predict, predict.lm

The following object is masked from 'package:base':

    print.default
```

```r
library(performance)
```

```
Warning: package 'performance' was built under R version 4.4.3
```

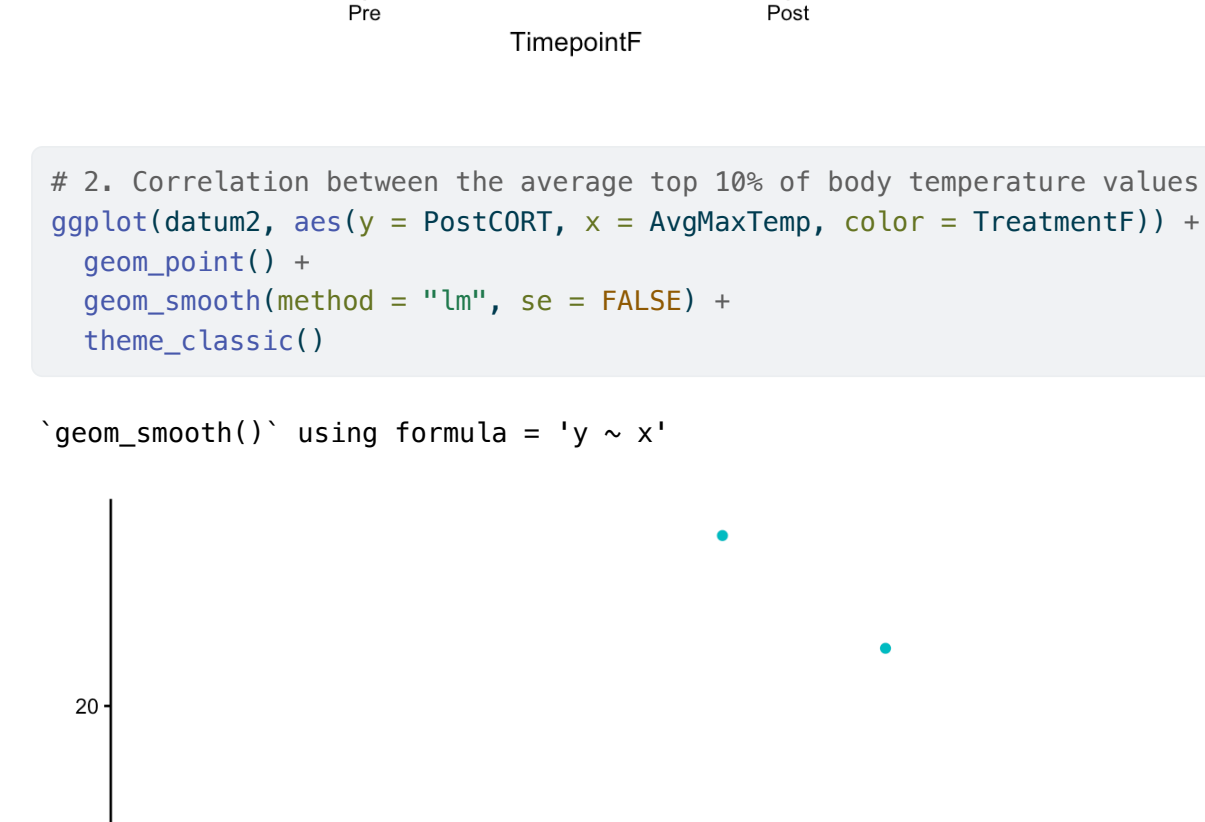
```r
library(see)
library(patchwork)
```

```

Attaching package: 'patchwork'

The following object is masked from 'package:cowplot':

    align_plots

The following object is masked from 'package:MASS':

    area
```

```r
# 2.
setwd("/Users/ben/Desktop/Blerd/Data Analysis")

# 3. Load your data
datum = read.csv("Blerd.csv")
datumlong = read.csv("BlerdData_long.csv")

# 4. Ensure categorical variables are properly set
datum$SexF = as.factor (datum$Sex)
datum$TreatmentF = as.factor(datum$Treatment)
datum$TimeF = as.factor(datum$Time)
datum$IncubatorF = as.factor (datum$Incubator)
datum$SideF = as.factor (datum$Side)
datum$PostCORT = as.numeric (datum$PostCORT)
datum$PreCORT = as.numeric (datum$PreCORT)


datumlong$SexF = as.factor (datumlong$Sex)
datumlong$TreatmentF = as.factor(datumlong$Treatment)
datumlong$TimepointF = as.factor(datumlong$Timepoint)
datumlong$CORT = as.numeric (datumlong$CORT)
datumlong$TimepointF = factor(datumlong$TimepointF, levels = c("Pre", "Post"))

# 5. Subset data so there are no dead individuals included
datum1 = datum %>%
  filter(!Died %in% c("Died", "PIT.Death"))

datumlong1 = datumlong %>%
  filter(!Died %in% c("Died", "PIT.Death"))

# 6. Take out any of the blank cells that cause problems when graphing
datum2 = datum1 %>%
  filter(PostCORT != "", PostCORT != "Dead", AvgMaxTemp != "", AvgMaxTemp != "Dead")
datum2$TreatmentF = as.factor(datum2$Treatment)

datumlong2 = datumlong1 %>%
  filter(CORT != "", Damage != "Dead")
```

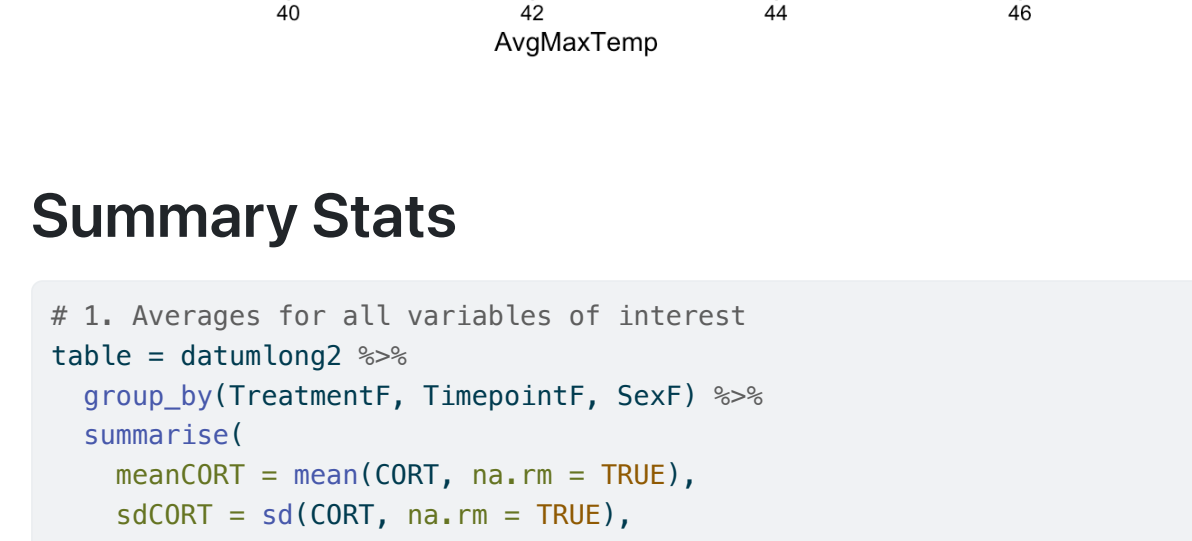
Initial Data Viz

```r
# 1. CORT across timepoints and relationship to treatment and sex
ggplot(datumlong2, aes(x = TimepointF, y = CORT, color = SexF, shape = TreatmentF)) +
  geom_point(position = position_jitterdodge(
      jitter.width = 0.2,
      dodge.width = 0.5 )) +
  theme_classic()
```



```r
# 2. Correlation between the average top 10% of body temperature values and post CORT
ggplot(datum2, aes(y = PostCORT, x = AvgMaxTemp, color = TreatmentF)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  theme_classic()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Summary Stats

```r
# 1. Averages for all variables of interest
table = datumlong2 %>%
  group_by(TreatmentF, TimepointF, SexF) %>%
  summarise(
    meanCORT = mean(CORT, na.rm = TRUE),
    sdCORT = sd(CORT, na.rm = TRUE),
    n = n(),
    .groups = "drop"
  )

print(table)
```

```
  TreatmentF TimepointF SexF meanCORT    sdCORT  n
1    Control        Pre    F 1.134822 1.0845040 19
2    Control        Pre    M 1.721281 1.6457407 15
3    Control       Post    F 2.730417 2.5204323 18
4    Control       Post    M 2.274688 2.5776196 18
5       Heat        Pre    F 1.339933 1.1150270 22
6       Heat        Pre    M 1.295712 0.9501043 19
7       Heat       Post    F 5.601528 7.1399297 19
8       Heat       Post    M 4.385074 2.9734675 15
```

Analysis - Glucocorticoids across time points

```r
# 1. Running the models

  #Start with global linear mixed effects model to assess damage levels
results = lmer(CORT ~ TreatmentF * SexF * TimepointF + (1|ID),
               data = datumlong2, na.action = na.omit, REML = FALSE)
summary(results)
```

```
Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
  method [lmerModLmerTest]
Formula: CORT ~ TreatmentF * SexF * TimepointF + (1 | ID)
   Data: datumlong2

     AIC      BIC   logLik -2*log(L)  df.resid
   756.2    786.0   -368.1     736.2       135

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.7648 -0.3637 -0.1790  0.1623  6.5798

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.577    0.7596
 Residual             8.825    2.9707
Number of obs: 145, groups:  ID, 81

Fixed effects:
                                    Estimate Std. Error       df t value
(Intercept)                           1.1348     0.7035 144.6052   1.613
TreatmentFHeat                        0.1990     0.9603 144.6399   0.207
SexFM                                 0.5743     1.0588 144.7846   0.542
TimepointFPost                        1.5932     0.9779  76.5663   1.629
TreatmentFHeat:SexFM                 -0.6186     1.4294 144.7413  -0.433
TreatmentFHeat:TimepointFPost         2.6670     1.3522  79.9705   1.972
SexFM:TimepointFPost                 -1.0334     1.4300  81.5970  -0.723
TreatmentFHeat:SexFM:TimepointFPost  -0.1176     1.9954  83.8422  -0.059
                                    Pr(>|t|)
(Intercept)                            0.109
TreatmentFHeat                         0.836
SexFM                                  0.588
TimepointFPost                         0.107
TreatmentFHeat:SexFM                   0.666
TreatmentFHeat:TimepointFPost          0.052 .
SexFM:TimepointFPost                   0.472
TreatmentFHeat:SexFM:TimepointFPost    0.953
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) TrtmFH SexFM  TmpnFP TrFH:SFM TFH:TF SFM:TF
TreatmntFHt -0.733
SexFM       -0.664  0.487
TimepntFPst -0.675  0.495  0.449
TrtmnFH:SFM  0.492 -0.672 -0.741 -0.332
TrtmnFH:TFP  0.488 -0.668 -0.324 -0.723  0.449
SxFM:TmpnFP  0.462 -0.338 -0.701 -0.684  0.519    0.495
TFH:SFM:TFP -0.331  0.452  0.502  0.490 -0.677   -0.678 -0.717
```

```r
Anova(results, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: CORT
                            Chisq Df Pr(>Chisq)
(Intercept)                2.6025  1    0.10670
TreatmentF                 0.0430  1    0.83581
SexF                       0.2942  1    0.58755
TimepointF                 2.6542  1    0.10328
TreatmentF:SexF            0.1873  1    0.66516
TreatmentF:TimepointF      3.8900  1    0.04857 *
SexF:TimepointF            0.5222  1    0.46990
TreatmentF:SexF:TimepointF 0.0035  1    0.95299
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# 2. The three way interaction was not significant, so it was removed and all two way in
results2 = lmer(CORT ~ TreatmentF * SexF + TreatmentF * TimepointF +
                  SexF * TimepointF + (1|ID), data = datumlong2, na.action = na.omit,
summary(results2)
```

```
Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
  method [lmerModLmerTest]
Formula:
CORT ~ TreatmentF * SexF + TreatmentF * TimepointF + SexF * TimepointF +
    (1 | ID)
   Data: datumlong2

     AIC      BIC   logLik -2*log(L)  df.resid
   754.2    781.0   -368.1     736.2       136

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.7602 -0.3590 -0.1795  0.1670  6.5842

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.5776   0.760
 Residual             8.8247   2.971
Number of obs: 145, groups:  ID, 81

Fixed effects:
                              Estimate Std. Error       df t value Pr(>|t|)
(Intercept)                     1.1211     0.6638 139.8148   1.689   0.0935 .
TreatmentFHeat                  0.2246     0.8564 130.7093   0.262   0.7935
SexFM                           0.6056     0.9157 130.1257   0.661   0.5095
TimepointFPost                  1.6214     0.8524  78.8016   1.902   0.0608 .
TreatmentFHeat:SexFM           -0.6756     1.0526  84.9215  -0.642   0.5227
TreatmentFHeat:TimepointFPost   2.6130     0.9943  83.2470   2.628   0.0102 *
SexFM:TimepointFPost           -1.0938     0.9973  83.9680  -1.097   0.2759
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) TrtmFH SexFM  TmpnFP TFH:SF TFH:TF
TreatmntFHt -0.693
SexFM       -0.610  0.336
TimepntFPst -0.624  0.351  0.269
TrtmnFH:SFM  0.386 -0.557 -0.630 -0.001
TrtmnFH:TFP  0.380 -0.550  0.025 -0.610 -0.018
SxFM:TmpnFP  0.341 -0.022 -0.565 -0.547  0.066  0.017
```

```r
Anova(results2, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: CORT
                       Chisq Df Pr(>Chisq)
(Intercept)           2.8522  1   0.091252 .
TreatmentF            0.0688  1   0.793089
SexF                  0.4374  1   0.508370
TimepointF            3.6183  1   0.057146 .
TreatmentF:SexF       0.4120  1   0.520961
TreatmentF:TimepointF 6.9055  1   0.008593 **
SexF:TimepointF       1.2029  1   0.272737
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# 3. The Sex*Treatment interaction was not significant, so it was removed from the model
results3 = lmer(CORT ~ TimepointF * SexF + TreatmentF * TimepointF +
                  (1|ID), data = datumlong2, na.action = na.omit, REML = FALSE)
summary(results3)
```

```
Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
  method [lmerModLmerTest]
Formula: CORT ~ TimepointF * SexF + TreatmentF * TimepointF + (1 | ID)
   Data: datumlong2

     AIC      BIC   logLik -2*log(L)  df.resid
   752.6    776.4   -368.3     736.6       137

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.7108 -0.3436 -0.1369  0.1541  6.6088

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.6222   0.7888
 Residual             8.8090   2.9680
Number of obs: 145, groups:  ID, 81

Fixed effects:
                              Estimate Std. Error        df t value Pr(>|t|)
(Intercept)                    1.28544    0.61331 144.60494   2.096   0.0378 *
TimepointFPost                 1.62045    0.85176  78.93993   1.902   0.0608 .
SexFM                          0.23512    0.71239 144.69278   0.330   0.7418
TreatmentFHeat                -0.08163    0.71239 144.68983  -0.115   0.9089
TimepointFPost:SexFM          -1.05045    0.99449  84.15507  -1.056   0.2939
TimepointFPost:TreatmentFHeat  2.60178    0.99355  83.37894   2.619   0.0105 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) TmpnFP SexFM  TrtmFH TFP:SF
TimepntFPst -0.674
SexFM       -0.513  0.344
TreatmntFHt -0.623  0.421 -0.022
TmpntFP:SFM  0.342 -0.548 -0.673  0.018
TmpntFP:TFH  0.419 -0.610  0.017 -0.674  0.018
```

```r
Anova(results3, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: CORT
                       Chisq Df Pr(>Chisq)
(Intercept)           4.3928  1   0.036091 *
TimepointF            3.6194  1   0.057110 .
SexF                  0.1089  1   0.741370
TreatmentF            0.0131  1   0.908775
TimepointF:SexF       1.1157  1   0.290844
TimepointF:TreatmentF 6.8574  1   0.008827 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# 4. The Time point*Sex interaction was not significant, so it was removed from the mode
results4 = lmer(CORT ~ TreatmentF * TimepointF + SexF + (1|ID), data = datumlong2,
                na.action = na.omit, REML = FALSE)
summary(results4)
```

```
Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
  method [lmerModLmerTest]
Formula: CORT ~ TreatmentF * TimepointF + SexF + (1 | ID)
   Data: datumlong2

     AIC      BIC   logLik -2*log(L)  df.resid
   751.7    772.5   -368.9     737.7       138

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.6249 -0.3832 -0.1765  0.1888  6.6734

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.5917   0.7692
 Residual             8.9101   2.9850
Number of obs: 145, groups:  ID, 81

Fixed effects:
                              Estimate Std. Error        df t value Pr(>|t|)
(Intercept)                    1.50732    0.57833 138.64455   2.606   0.0102 *
TreatmentFHeat                -0.06833    0.71495 144.73552  -0.096   0.9240
TimepointFPost                 1.12714    0.71634  80.96306   1.573   0.1195
SexFM                         -0.27175    0.52778  85.01912  -0.515   0.6080
TreatmentFHeat:TimepointFPost  2.62079    0.99885  83.27057   2.624   0.0103 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) TrtmFH TmpnFP SexFM
TreatmntFHt -0.670
TimepntFPst -0.620  0.516
SexFM       -0.406 -0.014 -0.041
TrtmnFH:TFP  0.440 -0.676 -0.718  0.040
```

```r
Anova(results4, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: CORT
                       Chisq Df Pr(>Chisq)
(Intercept)           6.7928  1   0.009152 **
TreatmentF            0.0091  1   0.923855
TimepointF            2.4758  1   0.115612
SexF                  0.2651  1   0.606620
TreatmentF:TimepointF 6.8844  1   0.008695 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
#Use AIC to test which one of these models has the best fit
AIC(results, results2, results3, results4)
```

```
         df      AIC
results  10 756.1846
results2  9 754.1880
results3  8 752.5978
results4  7 751.7082
```

```r
#This is the global model, use 'results4' for testing the assumptions

results_final = lmer(CORT ~ TreatmentF * TimepointF + SexF + (1|ID), data = datumlong2,
                na.action = na.omit, REML = TRUE)
summary(results_final)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: CORT ~ TreatmentF * TimepointF + SexF + (1 | ID)
   Data: datumlong2

REML criterion at convergence: 735.1

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.5955 -0.3753 -0.1723  0.1782  6.5429

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.6679   0.8172
 Residual             9.1766   3.0293
Number of obs: 145, groups:  ID, 81

Fixed effects:
                              Estimate Std. Error        df t value Pr(>|t|)
(Intercept)                    1.50670    0.58894 133.35138   2.558   0.0116 *
TreatmentFHeat                -0.06836    0.72771 139.61703  -0.094   0.9253
TimepointFPost                 1.12732    0.72715  78.68742   1.550   0.1251
SexFM                         -0.27160    0.53843  81.77334  -0.504   0.6153
TreatmentFHeat:TimepointFPost  2.62139    1.01400  80.92242   2.585   0.0115 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) TrtmFH TmpnFP SexFM
TreatmntFHt -0.670
TimepntFPst -0.618  0.514
SexFM       -0.407 -0.014 -0.041
TrtmnFH:TFP  0.439 -0.674 -0.718  0.040
```

```r
Anova(results_final, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: CORT
                       Chisq Df Pr(>Chisq)
(Intercept)           6.5450  1   0.010518 *
TreatmentF            0.0088  1   0.925153
TimepointF            2.4035  1   0.121063
SexF                  0.2545  1   0.613959
TreatmentF:TimepointF 6.6832  1   0.009733 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

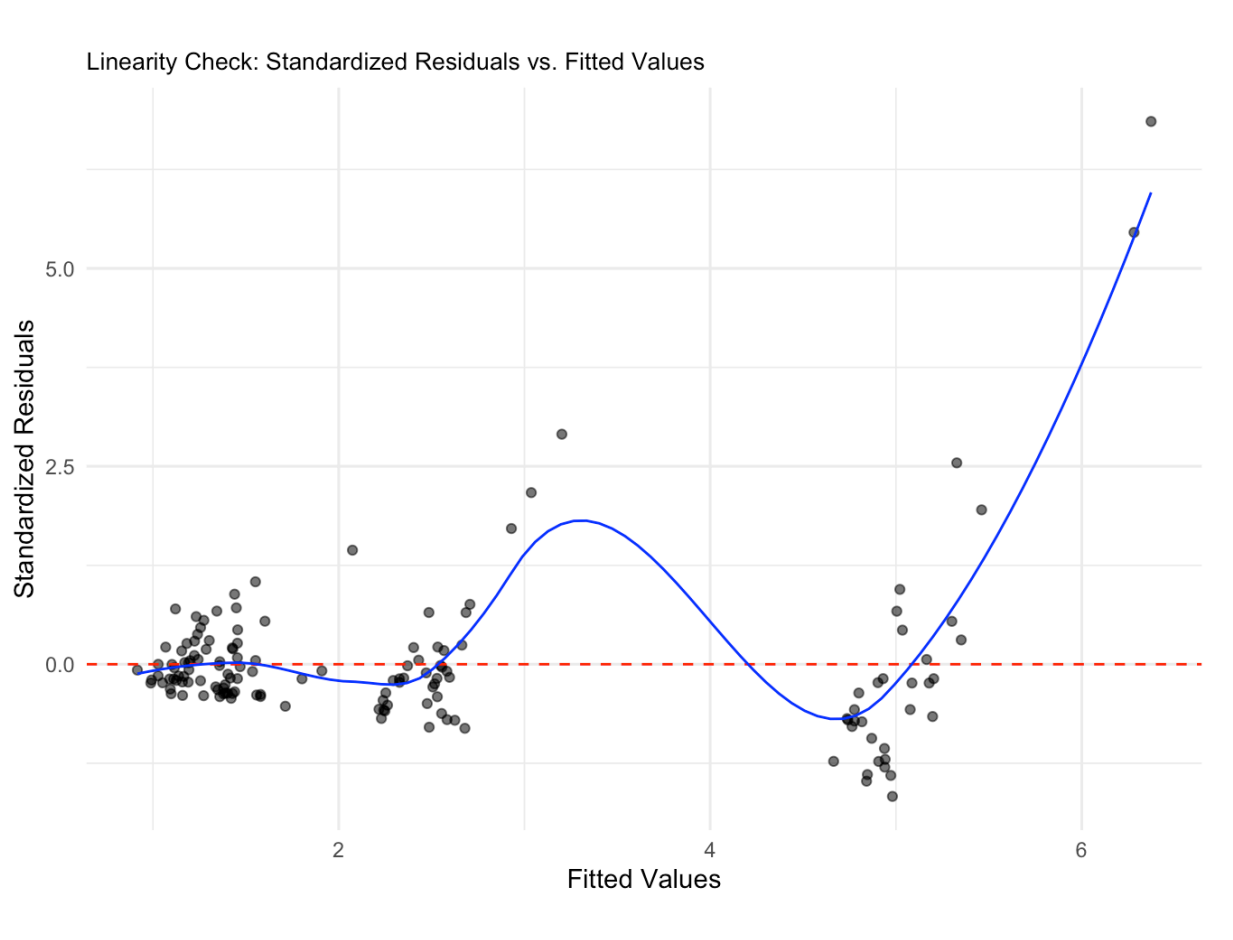
```r
#Assumptions of the model


# 1. Check the VIF of this model
vif(results4)
```

```
           TreatmentF            TimepointF                  SexF
             1.846740              2.070014              1.002288
TreatmentF:TimepointF
             2.795047
```

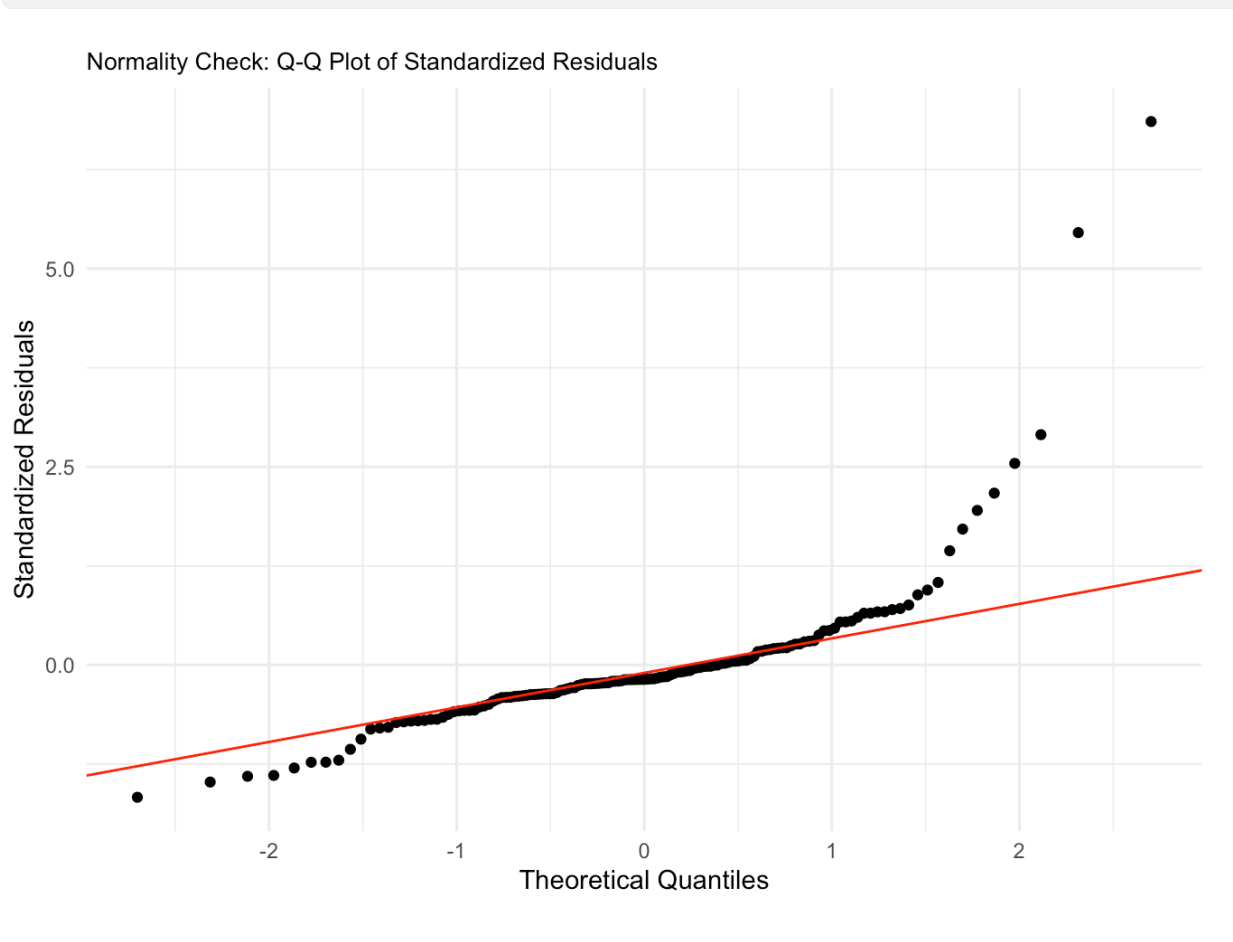
```r
  #VIF = 1: No multicollinearity
  #VIF between 1 and 5: Generally considered acceptable or moderate multicollinearity
  #VIF > 5 (or > 10): Indicates potentially high multicollinearity

# 2. Check other assumptions of a linear model

# --- Extract Residuals and Fitted Values ---
#1. Extract the data frame used by the model
datum_clean = model.frame(results4)
```

```
# 2. Calculate and assign the fitted and residual values to this clean data frame
datum_clean$fitted <- fitted(results4)
datum_clean$raw_resid <- residuals(results4)
datum_clean$std_resid <- datum_clean$raw_resid / sd(datum_clean$raw_resid, na.rm = TRUE)

# 3. Use the studentized residuals method from the 'lme4' utility functions
# Although 'rstandard' doesn't work, 'residuals' is robust.
# If you really need a standardized residual, use this function:
datum_clean$std_resid <- datum_clean$raw_resid / sd(datum_clean$raw_resid, na.rm = TRUE)


# --- 2. Create the Plots ---

# A. Linearity and Homoscedasticity Plot (Residuals vs. Fitted)
ggplot(datum_clean, aes(x = fitted, y = std_resid)) +
  geom_point(alpha = 0.6) +
  # Add the ideal reference line at y=0
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  # Add a LOESS smoothing line (red line in performance)
  geom_smooth(method = "loess", se = FALSE, color = "blue", linewidth = 0.5) +
  labs(
    title = "Linearity Check: Standardized Residuals vs. Fitted Values",
    x = "Fitted Values",
    y = "Standardized Residuals"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(size = 10))
```

```
`geom_smooth()` using formula = 'y ~ x'
```

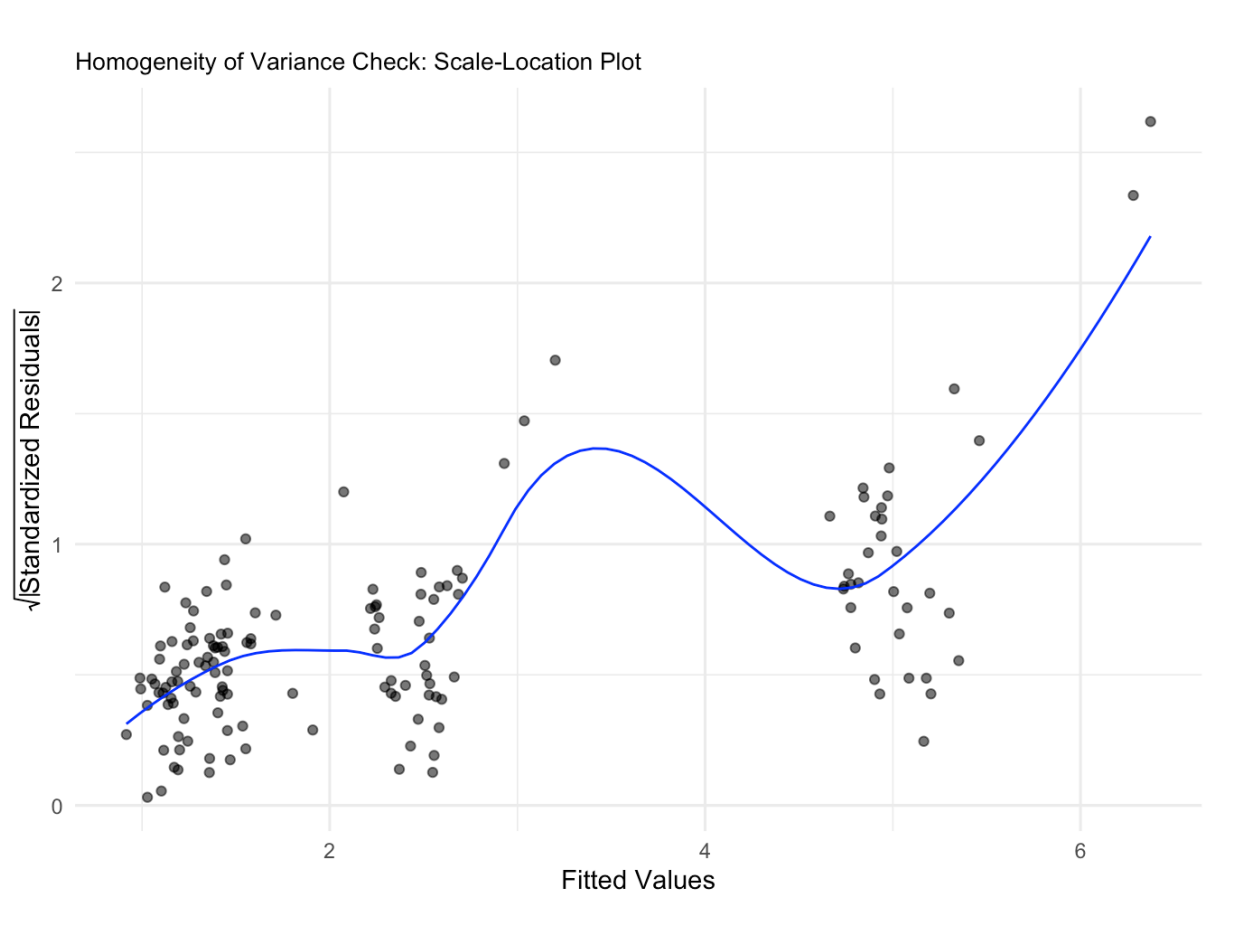


```
# B. Normality Check (Quantile-Quantile Plot)
ggplot(datum_clean, aes(sample = std_resid)) +
  # Create the Q-Q plot
  stat_qq() +
  # Add the reference line (the ideal normal distribution line)
  stat_qq_line(color = "red") +
  labs(
    title = "Normality Check: Q-Q Plot of Standardized Residuals",
    x = "Theoretical Quantiles",
    y = "Standardized Residuals"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(size = 10))
```

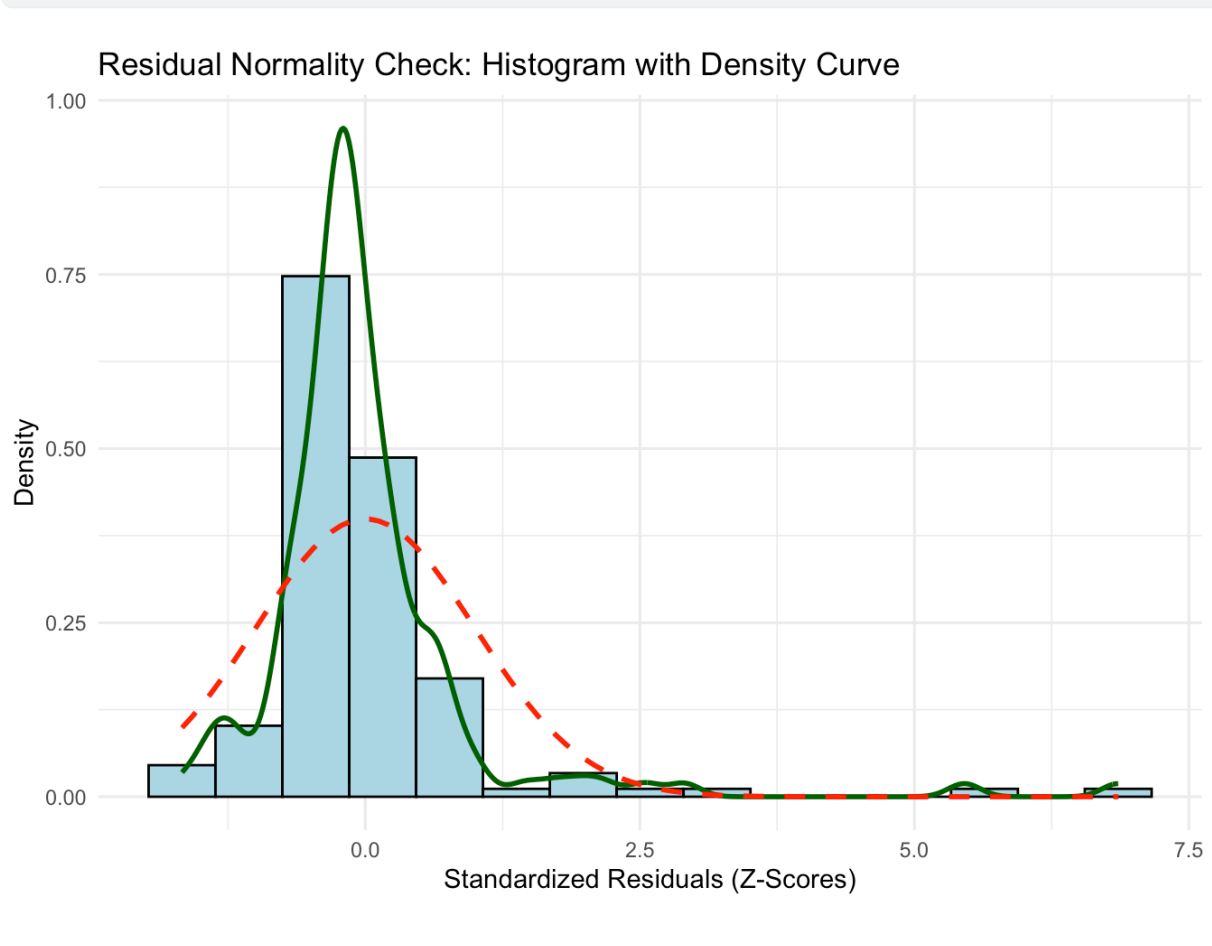


```
# C. Scale-Location Plot (Homogeneity of Variance Check)
# This uses the square root of the absolute standardized residuals
ggplot(datum_clean, aes(x = fitted, y = sqrt(abs(std_resid)))) +
  geom_point(alpha = 0.6) +
  # Add the ideal flat reference line
  geom_smooth(method = "loess", se = FALSE, color = "blue", linewidth = 0.5) +
  labs(
    title = "Homogeneity of Variance Check: Scale-Location Plot",
    x = "Fitted Values",
    y = expression(sqrt(abs("Standardized Residuals")))
  ) +
  theme_minimal() +
  theme(plot.title = element_text(size = 10))
```

```
`geom_smooth()` using formula = 'y ~ x'
```



```
#D. Histogram of residuals
# Calculate mean and standard deviation of residuals for the density curve
resid_mean <- mean(datum_clean$std_resid, na.rm = TRUE)
resid_sd <- sd(datum_clean$std_resid, na.rm = TRUE)

ggplot(datum_clean, aes(x = std_resid)) +
  # 1. Create the Histogram (using density scale)
  geom_histogram(aes(y = after_stat(density)),
                 bins = 15, # Adjust bin count for best visualization
                 color = "black",
                 fill = "lightblue") +

  # 2. Overlay the Density of the Residuals (Green Line)
  geom_density(color = "darkgreen", linewidth = 1) +

  # 3. Overlay the Theoretical Normal Density Curve (Red Dashed Line)
  stat_function(fun = dnorm,
                args = list(mean = resid_mean, sd = resid_sd),
                color = "red",
                linewidth = 1,
                linetype = "dashed") +

  labs(
    title = "Residual Normality Check: Histogram with Density Curve",
    x = "Standardized Residuals (Z-Scores)",
    y = "Density"
  ) +
  theme_minimal()
```



Conclusions: I do not love the way the residuals look, so I would like to try a gamma distribution

Next steps: Try gamma distribution glmer

```
# 1. Running the models

  #Start with global linear mixed effects model to assess damage levels
results = glmer(CORT ~ TreatmentF * SexF * TimepointF + (1|ID),
               data = datumlong2, na.action = na.omit,
               family = Gamma(link = "log"))
summary(results)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: Gamma  ( log )
Formula: CORT ~ TreatmentF * SexF * TimepointF + (1 | ID)
   Data: datumlong2

     AIC       BIC    logLik -2*log(L)  df.resid
   514.2     544.0    -247.1     494.2       135

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.3460 -0.6177 -0.0382  0.4735  3.1921

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.2615   0.5113
 Residual             0.5296   0.7277
Number of obs: 145, groups:  ID, 81

Fixed effects:
                                    Estimate Std. Error t value Pr(>|z|)
(Intercept)                         -0.05189    0.22699  -0.229 0.819196
TreatmentFHeat                       0.11236    0.30775   0.365 0.715028
SexFM                                0.26220    0.33376   0.786 0.432108
TimepointFPost                       0.93274    0.25703   3.629 0.000285 ***
TreatmentFHeat:SexFM                -0.21345    0.45288  -0.471 0.637413
TreatmentFHeat:TimepointFPost        0.29967    0.35773   0.838 0.402212
SexFM:TimepointFPost                -0.67288    0.36230  -1.857 0.063279 .
TreatmentFHeat:SexFM:TimepointFPost  0.81153    0.52133   1.557 0.119556
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) TrtmFH SexFM  TmpnFP TrFH:SFM TFH:TF SFM:TF
TreatmntFHt -0.721
SexFM       -0.663  0.500
TimepntFPst -0.559  0.404  0.372
TrtmnFH:SFM  0.484 -0.683 -0.740 -0.271
TrtmnFH:TFP  0.409 -0.530 -0.262 -0.722  0.355
SxFM:TmpnFP  0.398 -0.286 -0.567 -0.710  0.416    0.513
TFH:SFM:TFP -0.290  0.358  0.385  0.500 -0.527   -0.695 -0.701
```

```
Anova(results, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: CORT
                            Chisq Df Pr(>Chisq)
(Intercept)                0.0522  1  0.8191958
TreatmentF                 0.1333  1  0.7150279
SexF                       0.6172  1  0.4321078
TimepointF                13.1689  1  0.0002846 ***
TreatmentF:SexF            0.2221  1  0.6374132
TreatmentF:TimepointF      0.7017  1  0.4022118
SexF:TimepointF            3.4493  1  0.0632786 .
TreatmentF:SexF:TimepointF  2.4231  1  0.1195564
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# 2. The three way interaction was not significant, so it was removed and all two way in
results2 = glmer(CORT ~ TreatmentF * SexF + TreatmentF * TimepointF +
                SexF * TimepointF + (1|ID), data = datumlong2, na.action = na.omit,
                family = Gamma(link = "log"))
summary(results2)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: Gamma  ( log )
Formula:
CORT ~ TreatmentF * SexF + TreatmentF * TimepointF + SexF * TimepointF +
    (1 | ID)
   Data: datumlong2

     AIC       BIC    logLik -2*log(L)  df.resid
   514.6     541.4    -248.3     496.6       136

Scaled residuals:
     Min       1Q   Median       3Q      Max
-1.32488 -0.67035 -0.08887  0.47023  3.04463

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.2375   0.4874
 Residual             0.5459   0.7388
Number of obs: 145, groups:  ID, 81

Fixed effects:
                              Estimate Std. Error t value Pr(>|z|)
(Intercept)                    0.05665    0.21924   0.258  0.79609
TreatmentFHeat                -0.05728    0.28486  -0.201  0.84064
SexFM                          0.06664    0.30373   0.219  0.82633
TimepointFPost                 0.73213    0.22731   3.221  0.00128 **
TreatmentFHeat:SexFM           0.15041    0.37843   0.397  0.69103
TreatmentFHeat:TimepointFPost  0.68856    0.26209   2.627  0.00861 **
SexFM:TimepointFPost          -0.27889    0.26309  -1.060  0.28912
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) TrtmFH SexFM  TmpnFP TFH:SF TFH:TF
TreatmntFHt -0.698
SexFM       -0.635  0.442
TimepntFPst -0.542  0.321  0.269
TrtmnFH:SFM  0.404 -0.621 -0.679 -0.039
TrtmnFH:TFP  0.326 -0.433 -0.023 -0.599 -0.017
SxFM:TmpnFP  0.310 -0.082 -0.463 -0.584  0.074  0.048
```

```
Anova(results2, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: CORT
                        Chisq Df Pr(>Chisq)
(Intercept)            0.0668  1   0.796087
TreatmentF             0.0404  1   0.840644
SexF                   0.0481  1   0.826328
TimepointF            10.3740  1   0.001278 **
TreatmentF:SexF        0.1580  1   0.691029
TreatmentF:TimepointF  6.9023  1   0.008608 **
SexF:TimepointF        1.1237  1   0.289123
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# 3. The Sex*Treatment interaction was not significant, so it was removed from the model
results3 = glmer(CORT ~ TimepointF * SexF + TreatmentF * TimepointF +
                (1|ID), data = datumlong2, na.action = na.omit,
                family = Gamma(link = "log"))
summary(results3)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: Gamma  ( log )
Formula: CORT ~ TimepointF * SexF + TreatmentF * TimepointF + (1 | ID)
   Data: datumlong2

     AIC       BIC    logLik -2*log(L)  df.resid
   512.7     536.6    -248.4     496.7       137

Scaled residuals:
     Min       1Q   Median       3Q      Max
-1.32329 -0.66865 -0.09208  0.50179  3.00478

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.2307   0.4803
 Residual             0.5486   0.7407
Number of obs: 145, groups:  ID, 81

Fixed effects:
                              Estimate Std. Error t value Pr(>|z|)
(Intercept)                     0.0213     0.1998   0.107  0.91510
TimepointFPost                  0.7357     0.2268   3.243  0.00118 **
SexFM                           0.1487     0.2221   0.670  0.50299
TreatmentFHeat                  0.0130     0.2224   0.058  0.95338
TimepointFPost:SexFM           -0.2865     0.2633  -1.088  0.27648
TimepointFPost:TreatmentFHeat   0.6904     0.2628   2.627  0.00861 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) TmpnFP SexFM  TrtmFH TFP:SF
TimepntFPst -0.575
SexFM       -0.533  0.329
TreatmntFHt -0.620  0.378  0.026
TmpntFP:SFM  0.302 -0.577 -0.568 -0.039
TmpntFP:TFH  0.362 -0.596 -0.039 -0.569  0.036
```

```
Anova(results3, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: CORT
                        Chisq Df Pr(>Chisq)
(Intercept)            0.0114  1   0.915097
TimepointF            10.5193  1   0.001181 **
SexF                   0.4486  1   0.502990
TreatmentF             0.0034  1   0.953375
TimepointF:SexF        1.1843  1   0.276476
TimepointF:TreatmentF  6.9017  1   0.008611 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# 4. The Time point*Sex interaction was not significant, so it was removed from the mode
results4 = glmer(CORT ~ TreatmentF * TimepointF + SexF + (1|ID), data = datumlong2,
                na.action = na.omit, family = Gamma(link = "log"))
summary(results4)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: Gamma  ( log )
Formula: CORT ~ TreatmentF * TimepointF + SexF + (1 | ID)
   Data: datumlong2

     AIC       BIC    logLik -2*log(L)  df.resid
   511.9     532.8    -249.0     497.9       138

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.3119 -0.6821 -0.1046  0.4494  3.1147

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.2387   0.4886
 Residual             0.5564   0.7459
Number of obs: 145, groups:  ID, 81

Fixed effects:
                              Estimate Std. Error t value Pr(>|z|)
(Intercept)                   0.090627   0.192524   0.471  0.63783
TreatmentFHeat                0.003175   0.223394   0.014  0.98866
TimepointFPost                0.592926   0.186363   3.182  0.00146 **
SexFM                         0.011470   0.183951   0.062  0.95028
TreatmentFHeat:TimepointFPost 0.701047   0.263451   2.661  0.00779 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) TrtmFH TmpnFP SexFM
TreatmntFHt -0.637
TimepntFPst -0.522  0.436
SexFM       -0.470  0.006  0.028
TrtmnFH:TFP  0.368 -0.567 -0.707 -0.023
```

```
Anova(results4, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: CORT
                        Chisq Df Pr(>Chisq)
(Intercept)            0.2216  1   0.637833
TreatmentF             0.0002  1   0.988659
TimepointF            10.1224  1   0.001465 **
SexF                   0.0039  1   0.950279
TreatmentF:TimepointF  7.0810  1   0.007791 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#Use AIC to test which one of these models has the best fit
AIC(results, results2, results3, results4)
```

```
         df      AIC
results  10 514.2218
results2  9 514.5878
results3  8 512.7485
results4  7 511.9336
```

```
#This is the global model, use 'results4' for testing the assumptions

results_final = glmer(CORT ~ TreatmentF * TimepointF + SexF + (1|ID), data = datumlong2,
                na.action = na.omit, family = Gamma(link = "log"))
summary(results_final)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: Gamma  ( log )
Formula: CORT ~ TreatmentF * TimepointF + SexF + (1 | ID)
   Data: datumlong2

     AIC       BIC    logLik -2*log(L)  df.resid
   511.9     532.8    -249.0     497.9       138

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.3119 -0.6821 -0.1046  0.4494  3.1147

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.2387   0.4886
 Residual             0.5564   0.7459
Number of obs: 145, groups:  ID, 81

Fixed effects:
                              Estimate Std. Error t value Pr(>|z|)
(Intercept)                   0.090627   0.192524   0.471  0.63783
TreatmentFHeat                0.003175   0.223394   0.014  0.98866
TimepointFPost                0.592926   0.186363   3.182  0.00146 **
SexFM                         0.011470   0.183951   0.062  0.95028
TreatmentFHeat:TimepointFPost 0.701047   0.263451   2.661  0.00779 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) TrtmFH TmpnFP SexFM
TreatmntFHt -0.637
TimepntFPst -0.522  0.436
SexFM       -0.470  0.006  0.028
TrtmnFH:TFP  0.368 -0.567 -0.707 -0.023
```

```
Anova(results_final, type = "III")
```

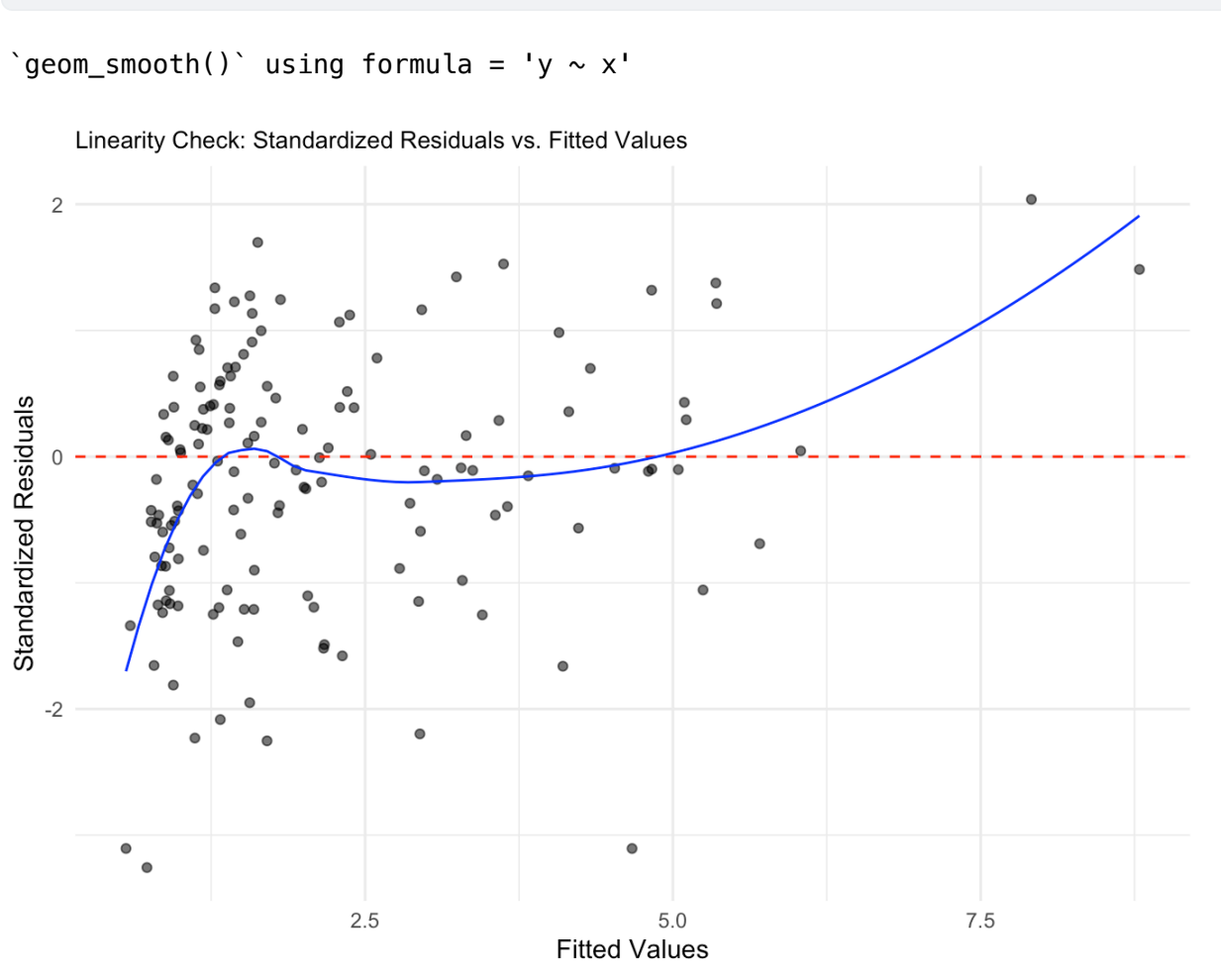
```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: CORT
                        Chisq Df Pr(>Chisq)
(Intercept)            0.2216  1   0.637833
TreatmentF             0.0002  1   0.988659
TimepointF            10.1224  1   0.001465 **
SexF                   0.0039  1   0.950279
TreatmentF:TimepointF  7.0810  1   0.007791 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#Assumptions of the model


# 1. Check the VIF of this model
vif(results4)
```

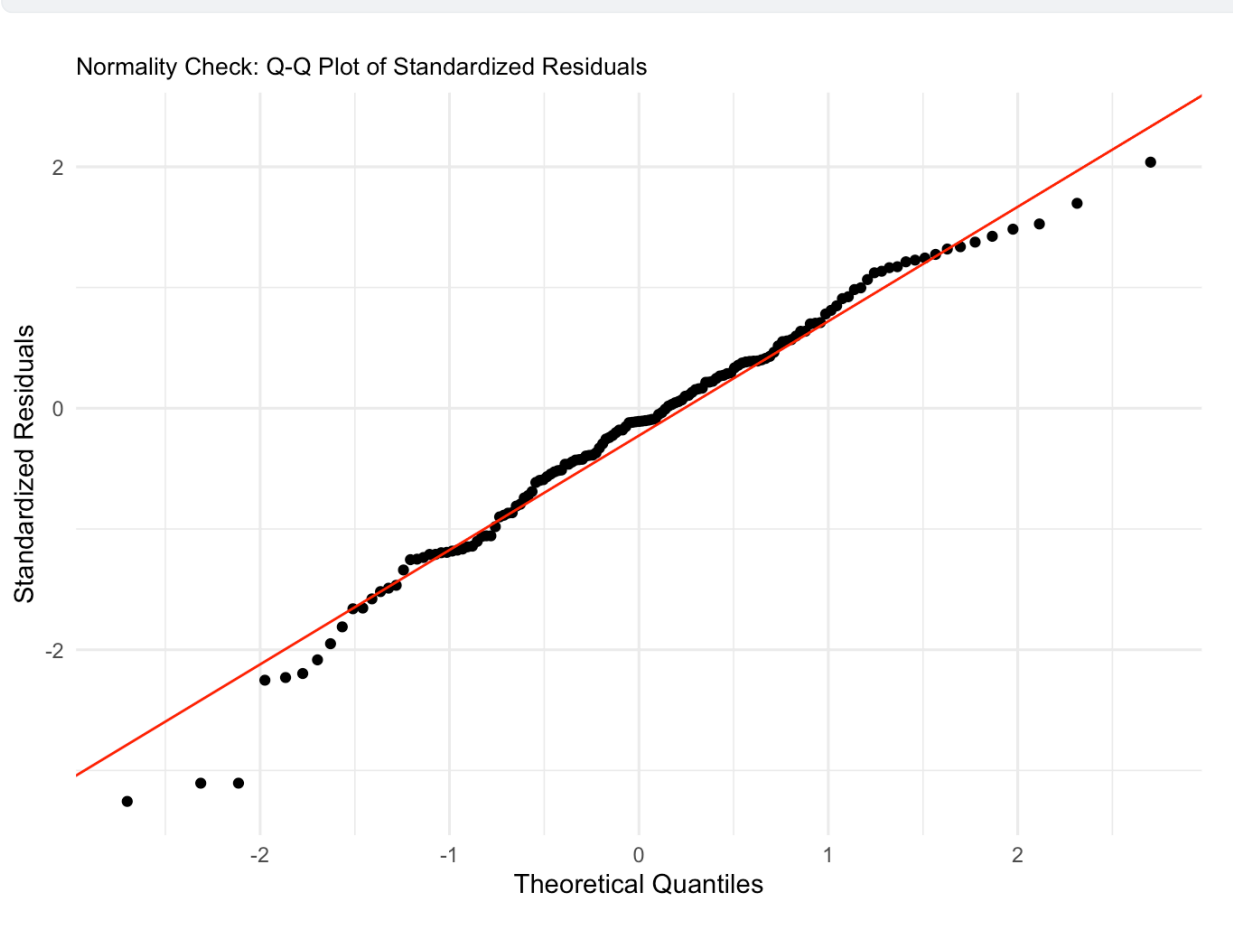
```
           TreatmentF            TimepointF                  SexF
             1.479921              2.008362              1.000889
TreatmentF:TimepointF
             2.397539
```

```
  #VIF = 1: No multicollinearity
  #VIF between 1 and 5: Generally considered acceptable or moderate multicollinearity
  #VIF > 5 (or > 10): Indicates potentially high multicollinearity

# 2. Check other assumptions of a linear model

# --- Extract Residuals and Fitted Values ---
#1. Extract the data frame used by the model
datum_clean = model.frame(results4)

# 2. Calculate and assign the fitted and residual values to this clean data frame
datum_clean$fitted <- fitted(results4)
datum_clean$raw_resid <- residuals(results4)
datum_clean$std_resid <- datum_clean$raw_resid / sd(datum_clean$raw_resid, na.rm = TRUE)

# 3. Use the studentized residuals method from the 'lme4' utility functions
# Although 'rstandard' doesn't work, 'residuals' is robust.
# If you really need a standardized residual, use this function:
datum_clean$std_resid <- datum_clean$raw_resid / sd(datum_clean$raw_resid, na.rm = TRUE)


# --- 2. Create the Plots ---

# A. Linearity and Homoscedasticity Plot (Residuals vs. Fitted)
ggplot(datum_clean, aes(x = fitted, y = std_resid)) +
  geom_point(alpha = 0.6) +
  # Add the ideal reference line at y=0
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  # Add a LOESS smoothing line (red line in performance)
  geom_smooth(method = "loess", se = FALSE, color = "blue", linewidth = 0.5) +
  labs(
    title = "Linearity Check: Standardized Residuals vs. Fitted Values",
    x = "Fitted Values",
    y = "Standardized Residuals"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(size = 10))
```

```
`geom_smooth()` using formula = 'y ~ x'
```

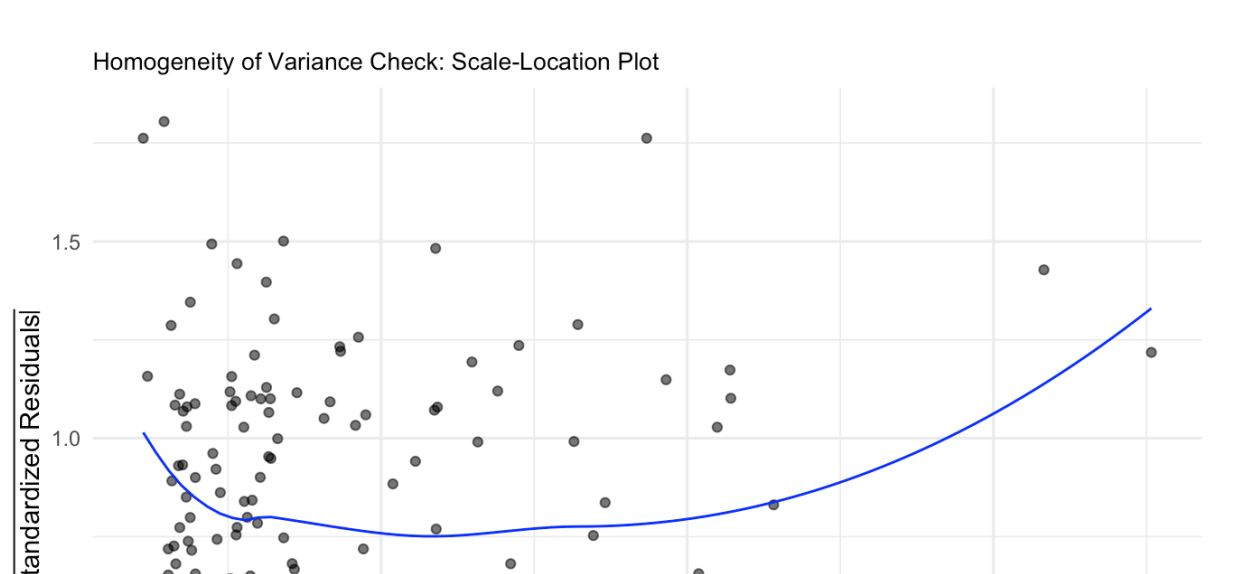


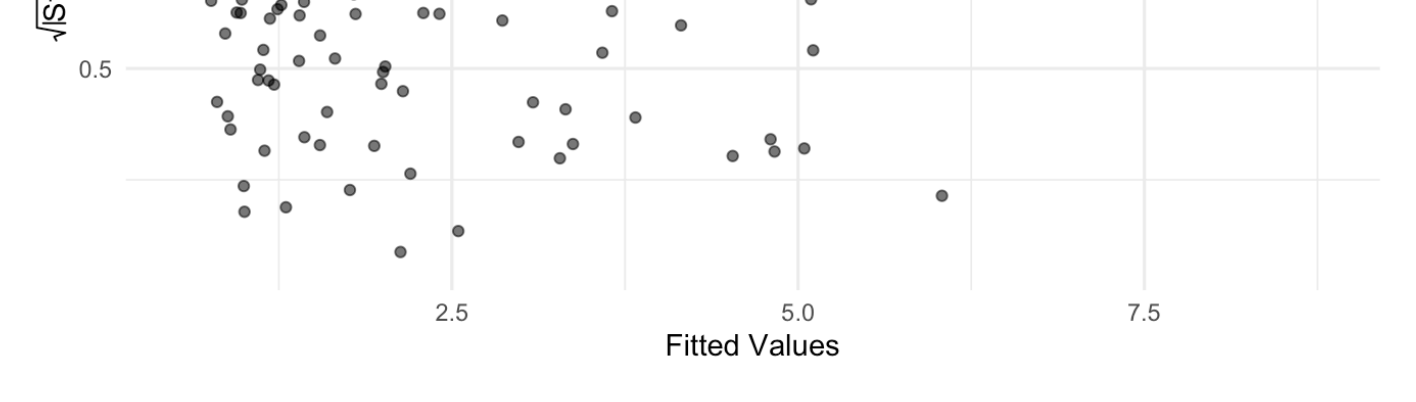
```
# B. Normality Check (Quantile-Quantile Plot)
ggplot(datum_clean, aes(sample = std_resid)) +
  # Create the Q-Q plot
  stat_qq() +
  # Add the reference line (the ideal normal distribution line)
  stat_qq_line(color = "red") +
  labs(
    title = "Normality Check: Q-Q Plot of Standardized Residuals",
    x = "Theoretical Quantiles",
    y = "Standardized Residuals"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(size = 10))
```



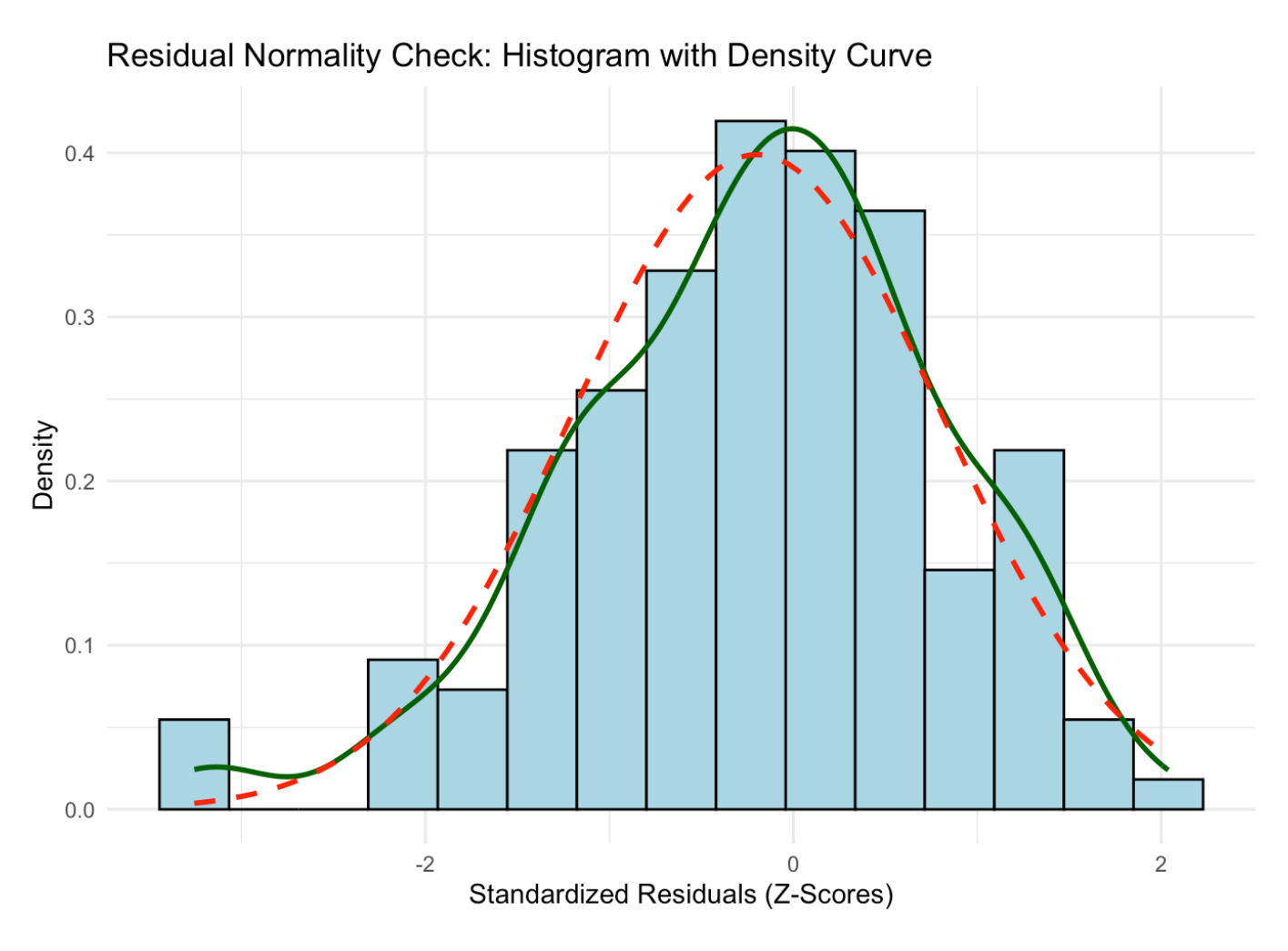
```
# C. Scale-Location Plot (Homogeneity of Variance Check)
# This uses the square root of the absolute standardized residuals
ggplot(datum_clean, aes(x = fitted, y = sqrt(abs(std_resid)))) +
  geom_point(alpha = 0.6) +
  # Add the ideal flat reference line
  geom_smooth(method = "loess", se = FALSE, color = "blue", linewidth = 0.5) +
  labs(
    title = "Homogeneity of Variance Check: Scale-Location Plot",
    x = "Fitted Values",
    y = expression(sqrt(abs("Standardized Residuals")))
  ) +
  theme_minimal() +
  theme(plot.title = element_text(size = 10))
```

```
`geom_smooth()` using formula = 'y ~ x'
```





```r
#D. Histogram of residuals
# Calculate mean and standard deviation of residuals for the density curve
resid_mean <- mean(datum_clean$std_resid, na.rm = TRUE)
resid_sd <- sd(datum_clean$std_resid, na.rm = TRUE)

ggplot(datum_clean, aes(x = std_resid)) +
  # 1. Create the Histogram (using density scale)
  geom_histogram(aes(y = after_stat(density)),
                 bins = 15, # Adjust bin count for best visualization
                 color = "black",
                 fill = "lightblue") +

  # 2. Overlay the Density of the Residuals (Green Line)
  geom_density(color = "darkgreen", linewidth = 1) +

  # 3. Overlay the Theoretical Normal Density Curve (Red Dashed Line)
  stat_function(fun = dnorm,
                args = list(mean = resid_mean, sd = resid_sd),
                color = "red",
                linewidth = 1,
                linetype = "dashed") +

  labs(
    title = "Residual Normality Check: Histogram with Density Curve",
    x = "Standardized Residuals (Z-Scores)",
    y = "Density"
  ) +
  theme_minimal()
```



```r
emms_interaction = emmeans(results_final, ~ TreatmentF * TimepointF)
print(emms_interaction)
```

```
 TreatmentF TimepointF emmean    SE  df asymp.LCL asymp.UCL
 Control    Pre        0.0964 0.170 Inf    -0.237     0.429
 Heat       Pre        0.0995 0.156 Inf    -0.206     0.405
 Control    Post       0.6893 0.165 Inf     0.367     1.012
 Heat       Post       1.3935 0.168 Inf     1.065     1.722

Results are averaged over the levels of: SexF 
Results are given on the log (not the response) scale. 
Confidence level used: 0.95 
```

```r
interaction_pairs = pairs(emms_interaction, adjust = "sidak")
print(interaction_pairs)
```

```
 contrast                   estimate    SE  df z.ratio p.value
 Control Pre - Heat Pre     -0.00318 0.223 Inf  -0.014  1.0000
 Control Pre - Control Post -0.59293 0.186 Inf  -3.182  0.0088
 Control Pre - Heat Post    -1.29715 0.233 Inf  -5.575  <.0001
 Heat Pre - Control Post    -0.58975 0.220 Inf  -2.683  0.0430
 Heat Pre - Heat Post       -1.29397 0.186 Inf  -6.947  <.0001
 Control Post - Heat Post   -0.70422 0.229 Inf  -3.072  0.0127

Results are averaged over the levels of: SexF 
Results are given on the log (not the response) scale. 
P value adjustment: sidak method for 6 tests 
```

```r
cld_output = cld(emms_interaction,
                 adjust = "sidak",
                 Letters = letters)

print(cld_output)
```

```
  TreatmentF TimepointF     emmean        SE  df  asymp.LCL asymp.UCL .group
1    Control        Pre 0.09636228 0.1699143 Inf -0.3268797 0.5196043    a  
2       Heat        Pre 0.09953772 0.1557457 Inf -0.2884116 0.4874870    a  
3    Control       Post 0.68928812 0.1645506 Inf  0.2794066 1.0991697     b 
4       Heat       Post 1.39351068 0.1677075 Inf  0.9757656 1.8112557      c
```

Analysis - Body Temperature and glucocorticoids

```r
# 1. Run Pearson's Correlation Test for both heated and control groups
pearson_result = cor.test(
  x = datum2$PostCORT,
  y = datum2$AvgMaxTemp,
  method = "pearson"
)
  # View the results
print(pearson_result)
```

```
Results of Hypothesis Test
--------------------------

Null Hypothesis:                 correlation = 0

Alternative Hypothesis:          True correlation is not equal to 0

Test Name:                       Pearson's product-moment correlation

Estimated Parameter(s):          cor = 0.3063535

Data:                            datum2$PostCORT and datum2$AvgMaxTemp

Test Statistic:                  t = 2.320732

Test Statistic Parameter:        df = 52

P-value:                         0.02425745

95% Confidence Interval:         LCL = 0.04204158
                                 UCL = 0.53059007
```

```r
# 2.  Run Pearson's Correlation Test heated group
datumHeated = datum2 %>%
 filter(TreatmentF == "Heat")

pearson_resultHeat = cor.test(
  x = datumHeated$PostCORT,
  y = datumHeated$AvgMaxTemp,
  method = "pearson"
)
  # View the results
print(pearson_resultHeat)
```

```
Results of Hypothesis Test
--------------------------

Null Hypothesis:                 correlation = 0

Alternative Hypothesis:          True correlation is not equal to 0

Test Name:                       Pearson's product-moment correlation

Estimated Parameter(s):          cor = 0.1358109

Data:                            datumHeated$PostCORT and datumHeated$AvgMaxTemp

Test Statistic:                  t = 0.6854047

Test Statistic Parameter:        df = 25

P-value:                         0.4993953

95% Confidence Interval:         LCL = -0.2574923
                                 UCL =  0.4905096
```

```r
# 2.  Run Pearson's Correlation Test control group
datumControl = datum2 %>%
 filter(TreatmentF == "Control")

pearson_resultControl = cor.test(
  x = datumControl$PostCORT,
  y = datumControl$AvgMaxTemp,
  method = "pearson"
)
  # View the results
print(pearson_resultControl)
```

```
Results of Hypothesis Test
--------------------------

Null Hypothesis:                 correlation = 0

Alternative Hypothesis:          True correlation is not equal to 0

Test Name:                       Pearson's product-moment correlation

Estimated Parameter(s):          cor = 0.01357196

Data:                            datumControl$PostCORT and datumControl$AvgMaxTemp

Test Statistic:                  t = 0.06786604

Test Statistic Parameter:        df = 25

P-value:                         0.9464322

95% Confidence Interval:         LCL = -0.3683417
                                 UCL =  0.3915664
```

Manuscript Graph

```r
# 1. Add the means and 95% CIs to the dataframe for the variables we are interested in

datumlong2_with_summary = datumlong2 %>%

  # 1. Group the data by the interaction terms
  group_by(TreatmentF, TimepointF) %>%

  # 2. Calculate the group-level statistics (Mean, CI, etc.)
  mutate(
    N_group = n(),
    Mean_CORT = mean(CORT, na.rm = TRUE),
    SD_CORT = sd(CORT, na.rm = TRUE),
    SE_CORT = SD_CORT / sqrt(N_group),
    CI_CORT = 1.96 * SE_CORT,
    CI_Upper_Group = Mean_CORT + CI_CORT,
    CI_Lower_Group = Mean_CORT - CI_CORT
  ) %>%
  ungroup()
  #View the dataframe that was created
head(datumlong2_with_summary)
```

```
# A tibble: 6 × 35
     ID Sex   Treatment Incubator Side  Timepoint Died  Date     Mass Glucose
  <int> <chr> <chr>     <chr>     <chr> <chr>     <chr> <chr>   <dbl> <chr>  
1  4819 M     Control   Aphrodite Right Pre       No    3/5/25   15.6 198.5  
2  4820 M     Control   Zeus      Right Pre       No    3/7/25   14.1 191.5  
3  4823 M     Heat      Aphrodite Right Pre       No    3/20/25  14.4 239    
4  4824 M     Heat      Apollo    Left  Pre       No    3/8/25   14.5 226.5  
5  4825 M     Heat      Aphrodite Right Pre       No    3/9/25   13.3 168.5  
6  4826 M     Control   Apollo    Left  Pre       No    3/11/25  14.8 192    
# ℹ 25 more variables: Ketones <chr>, HCT <chr>, CORT <dbl>, Damage <dbl>,
#   X <lgl>, X.1 <lgl>, X.2 <lgl>, X.3 <lgl>, X.4 <lgl>, X.5 <lgl>, X.6 <lgl>,
#   X.7 <lgl>, X.8 <lgl>, X.9 <lgl>, X.10 <lgl>, SexF <fct>, TreatmentF <fct>,
#   TimepointF <fct>, N_group <int>, Mean_CORT <dbl>, SD_CORT <dbl>,
#   SE_CORT <dbl>, CI_CORT <dbl>, CI_Upper_Group <dbl>, CI_Lower_Group <dbl>
```

```r
# 2. Cleaning up data frame before plotting
  # Define the consistent dodge width and rename the data frame for plotting
DODGE_WIDTH = 0.4
plot_data = datumlong2_with_summary

  #Order the time points so that Pre and Post are graphed in the correct order on the pl
plot_data$TreatmentF = as.factor(plot_data$TreatmentF)
plot_data$TimepointF = factor(plot_data$TimepointF, levels = c("Pre", "Post"))

# 3. Plotting the graph

plot = ggplot(data = plot_data, aes(x = TimepointF, y = CORT, color = TreatmentF)) +
  geom_jitter(
    size = 3.5,
    alpha = 0.5,
    position = position_jitterdodge(jitter.width = 0.2, dodge.width = DODGE_WIDTH)
  ) +

  geom_point(
    aes(y = Mean_CORT, group = TreatmentF),
    size = 7.5,
    stroke = 1,
    color = "black",
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  geom_errorbar(
    aes(ymin = CI_Lower_Group, ymax = CI_Upper_Group, group = TreatmentF),
    width = 0.2,
    linewidth = 1,
    color = "black",
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  geom_line(
    aes(y = Mean_CORT, group = TreatmentF),
    linewidth = 1.5,
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  scale_color_manual(
    values = c(
      "Control" = "#AAB8C2",
      "Heat" = "firebrick4"
    )
  ) +

  labs(
    x = "Time Point",
    y = "Corticosterone (ng/mL)",
    color = "Treatment Group"
  ) +
  theme_classic() +

  theme(
    axis.title = element_text(face = "bold", size = 30),
    axis.text = element_text(size = 25),
    legend.justification = c(0.5, 0.5),
    legend.title = element_text(size = 28),
    legend.text = element_text(size = 25),
    strip.text = element_text(size = 20),
    panel.background = element_rect(fill = "transparent", color = NA),
    plot.background = element_rect(fill = "transparent", color = NA),
    legend.background = element_rect(fill = "transparent", color = NA),
    legend.box.background = element_rect(fill = "transparent", color = NA),
    axis.line = element_line(linewidth = 1.5, colour = "black")
  )


# 4. Adding signifigance and sample size

plot1 = plot +
  annotate(
    "text",
    x = c(.9, 1.1, 1.9, 2.1),
    y = c(6.26 + 5, 3.51 + 5, 11.64 + 5, 26.29 + 5),
    label = c("a", "a", "b", "c"),
    size = 10,
    color = "black") +

   # Add a new layer for the sample sizes
  annotate(
    "text",
    x = c(.9, 1.1, 1.9, 2.1),
    # Adjust the y-coordinates to be slightly lower
    y = c(6.26 + 2.5, 3.51 + 2.5, 11.64 + 2.5, 26.29 + 2.5),
    # Add your sample sizes here
    label = c("(34)", "(36)", "(41)", "(34)"),
    size = 7.5, # Make the font size slightly smaller
    color = "black")

# 5. Save the graph
ggsave(
  filename = "CORTPlot.png",
  plot = plot1,
  width = 11.04,
  height = 6.49,
  units = "in",
  dpi = 300,
  bg = "transparent"
)
```

Manuscript Graph - With sex

```r
# 1. Add the means and 95% CIs to the dataframe for the variables we are interested in

datumlong2_with_summary = datumlong2 %>%

  # 1. Group the data by the interaction terms
  group_by(TreatmentF, TimepointF) %>%

  # 2. Calculate the group-level statistics (Mean, CI, etc.)
  mutate(
    N_group = n(),
    Mean_CORT = mean(CORT, na.rm = TRUE),
    SD_CORT = sd(CORT, na.rm = TRUE),
    SE_CORT = SD_CORT / sqrt(N_group),
    CI_CORT = 1.96 * SE_CORT,
    CI_Upper_Group = Mean_CORT + CI_CORT,
    CI_Lower_Group = Mean_CORT - CI_CORT
  ) %>%
  ungroup()
  #View the dataframe that was created
head(datumlong2_with_summary)
```

```
# A tibble: 6 × 35
     ID Sex   Treatment Incubator Side  Timepoint Died  Date     Mass Glucose
  <int> <chr> <chr>     <chr>     <chr> <chr>     <chr> <chr>   <dbl> <chr>  
1  4819 M     Control   Aphrodite Right Pre       No    3/5/25   15.6 198.5  
2  4820 M     Control   Zeus      Right Pre       No    3/7/25   14.1 191.5  
3  4823 M     Heat      Aphrodite Right Pre       No    3/20/25  14.4 239    
4  4824 M     Heat      Apollo    Left  Pre       No    3/8/25   14.5 226.5  
5  4825 M     Heat      Aphrodite Right Pre       No    3/9/25   13.3 168.5  
6  4826 M     Control   Apollo    Left  Pre       No    3/11/25  14.8 192    
# ℹ 25 more variables: Ketones <chr>, HCT <chr>, CORT <dbl>, Damage <dbl>,
#   X <lgl>, X.1 <lgl>, X.2 <lgl>, X.3 <lgl>, X.4 <lgl>, X.5 <lgl>, X.6 <lgl>,
#   X.7 <lgl>, X.8 <lgl>, X.9 <lgl>, X.10 <lgl>, SexF <fct>, TreatmentF <fct>,
#   TimepointF <fct>, N_group <int>, Mean_CORT <dbl>, SD_CORT <dbl>,
#   SE_CORT <dbl>, CI_CORT <dbl>, CI_Upper_Group <dbl>, CI_Lower_Group <dbl>
```

```r
# 2. Cleaning up data frame before plotting
  # Define the consistent dodge width and rename the data frame for plotting
DODGE_WIDTH = 0.4
plot_data = datumlong2_with_summary

  #Order the time points so that Pre and Post are graphed in the correct order on the pl
plot_data$TreatmentF = as.factor(plot_data$TreatmentF)
plot_data$TimepointF = factor(plot_data$TimepointF, levels = c("Pre", "Post"))

# 3. Plotting the graph

plot = ggplot(data = plot_data, aes(x = TimepointF, y = CORT, color = TreatmentF)) +

geom_jitter(
  aes(shape = SexF, group = TreatmentF), # Group by Treatment ONLY
  size = 3.5,
  alpha = 0.5,
  position = position_jitterdodge(
    jitter.width = 0.2,
    dodge.width = DODGE_WIDTH
  )
) +

  geom_point(
    aes(y = Mean_CORT, group = TreatmentF),
    size = 7.5,
    shape = 18,
    stroke = 1,
    color = "black",
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  geom_errorbar(
    aes(ymin = CI_Lower_Group, ymax = CI_Upper_Group, group = TreatmentF),
    width = 0.2,
    linewidth = 1,
    color = "black",
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  geom_line(
    aes(y = Mean_CORT, group = TreatmentF),
    linewidth = 1.5,
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  scale_color_manual(
    values = c(
      "Control" = "#AAB8C2",
      "Heat" = "firebrick4"
    )
  ) +

  labs(
    x = "Time Point",
    y = "Corticosterone (ng/mL)",
    color = "Treatment Group"
  ) +
  guides(shape = "none") +

  theme_classic() +

  theme(
    axis.title = element_text(face = "bold", size = 30),
    axis.text = element_text(size = 25),
    legend.justification = c(0.5, 0.5),
    legend.title = element_text(size = 28),
    legend.text = element_text(size = 25),
    strip.text = element_text(size = 20),
    panel.background = element_rect(fill = "transparent", color = NA),
    plot.background = element_rect(fill = "transparent", color = NA),
    legend.background = element_rect(fill = "transparent", color = NA),
    legend.box.background = element_rect(fill = "transparent", color = NA),
    axis.line = element_line(linewidth = 1.5, colour = "black")
  )


# 4. Adding signifigance and sample size

plot1 = plot +
  annotate(
    "text",
    x = c(.9, 1.1, 1.9, 2.1),
    y = c(6.26 + 5, 3.51 + 5, 11.64 + 5, 26.29 + 5),
    label = c("a", "a", "b", "c"),
    size = 10,
    color = "black") +

   # Add a new layer for the sample sizes
  annotate(
    "text",
    x = c(.9, 1.1, 1.9, 2.1),
    # Adjust the y-coordinates to be slightly lower
    y = c(6.26 + 2.5, 3.51 + 2.5, 11.64 + 2.5, 26.29 + 2.5),
    # Add your sample sizes here
    label = c("(34)", "(36)", "(41)", "(34)"),
    size = 7.5, # Make the font size slightly smaller
    color = "black")

# 5. Save the graph
ggsave(
  filename = "CORTPlotSex.png",
  plot = plot1,
  width = 11.04,
  height = 6.49,
  units = "in",
  dpi = 300,
  bg = "transparent"
)
```